

The m -pancycle-connectivity of Basic WK-recursive Pyramids

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Abstract

The basic WK-recursive pyramid, denoted by $WKP_{(d,L)}$, is similar to pyramid networks, but is based on the WK-recursive network. A graph is *pancyclic* if it embeds cycles of all lengths ranging from 3 to N , where N is the order of the graph. A graph G is *m -pancycle-connected* if and only if any two nodes of G are in a common cycle of all lengths ranging from m to $|V(G)|$. This paper shows that $WKP_{(d,L)}$ is *pancyclic* and $(2^{L+1} - 2)$ -pancycle-connected.

1 Introduction

The interconnection network is important for determining the overall performance of a multiprocessor/multicomputer system. If the networks cannot provide adequate performance for a particular application, nodes will frequently be forced to wait for data arriving. Therefore, it is important to design an interconnection network with better performance. Let processes or cores be considered as nodes, and communication channels be considered as edges, then all problems on interconnection networks can be discussed on graphs.

Suppose that G is a graph. A path (cycle) in G is called a *Hamiltonian path* (*Hamiltonian cycle*) if it contains every node of G exactly once. G is called *Hamiltonian* if there is a Hamiltonian cycle in G , and it is called *Hamiltonian connected* if there is a Hamiltonian path between every two distinct nodes of G [2]. A graph is *pancyclic* if it embeds cycles of all lengths ranging from 3 to N , where N is the order of the graph [1]. A graph is *m -pancyclic* if it embeds cycles of all lengths ranging from m to N , where $3 \leq m \leq N$. Clearly, an *m -pancyclic* graph must be Hamiltonian. A graph

G is *m -pancycle-connected* if and only if any two nodes of G are in a common cycle of all lengths ranging from m to $|V(G)|$. A graph G with N vertices is *panconnected* if for any two distinct nodes x, y and for any integer $d(x, y) \leq l \leq N - 1$, there exists a path of length l in G connecting x and y , where $d(x, y)$ is the distance between x and y in G [3]. A graph G with N vertices is *m -path-panconnected* (*m -panconnected* for short) if for any two distinct nodes x, y and for any integer $m \leq l \leq N - 1$, there exists a path of length l in G connecting x and y .

The d dimension t amplitude WK-recursive network ($WK_{(d,t)}$, for short), proposed in 1992 by Fernandes [6], is a network that is recursively defined and is expandable to any level. This is also well-known networks in network computing, there are several researches on WK-recursive network ([4], [8], [9]). In fact, $WK_{(d,t)}$ is a general version of Hanoi graphs, i.e. the Hanoi graphs is isomorphic to $WK_{(3,t)}$. In 1993, Fernandes and Kanevsky proposed two hierarchical WK-recursive topologies, Hierarchical WK-Recursive Networks and WK-Recursive Pyramid Networks ($HWK_{(d,t,L)}$ and $WKP_{(d,t,L)}$, for short, respectively) [7]. They are constructed by taking difference size WK-recursive network as difference layers. [7] discussed about the orders sizes and connectivity of $HWK_{(d,t,L)}$ and $WKP_{(d,t,L)}$. In 2007, Fang et. al. proposed the definition of pancycle-connectivity of $WK_{(d,t)}$, and they show that $WK_{(d,t)}$ is $(5(2^{t-1}) - 2)$ -pancycle-connected for $n \geq 3$ [5]. In 2008, Razavi and Sarbazi-Azad claimed that a special graph, called WKP, is Hamiltonian and pancyclic, and also proved those properties [9]. In fact, WKP is a special version of $WKP_{(d,t,L)}$, where $d = 4$ and $t = 1$. In topology WKP, each node has exactly 4 children, and those children form a complete K_4 . In 2012, another special version of $WKP_{(d,t,L)}$, called basic WK-recursive pyramid, denoted by $WKP_{(d,L)}$, was proposed by Wang and Juan [10]. In $WKP_{(d,L)}$,

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each node has exactly d children and the n th layer is isomorphic to $WK_{(d,n)}$. Therefore, WKP is a special version of $WKP_{(d,L)}$, where $d = 4$, by the definition in [10].

In this paper, we will give some definitions and show some preliminaries in Section 2. Section 3 prove the main result that $WKP_{(d,L)}$ is pancyclic and $(2^{L+1} - 2)$ -pancycle-connected. Some conclusion are given in Section 4.

2 Definitions and Preliminaries

In this section, we review the definitions of $WK_{(d,t)}$, $WKP_{(d,t,L)}$ and $WKP_{(d,L)}$, and also show some preliminaries of them. At first, we review the definition of WK -recursive network.

Definition 1 [6] For $t \geq 1$, a radix- t WK -recursive network, denoted as $WK_{(d,t)}$, consists of a set of nodes $V(WK_{(d,t)}) = \{a_{t-1}a_{t-2}\dots a_1a_0 \mid 0 \leq a_i < d, 0 \leq i \leq t-1\}$. Each node $a_{t-1}a_{t-2}\dots a_1a_0$ is adjacent to 1) $a_{t-1}a_{t-2}\dots a_1b$, where $b \neq a_0$; and adjacent to 2) $a_{t-1}a_{t-2}\dots a_{j+1}b_1(b_0)^j$, if $a_j \neq a_{j-1}$ and $a_{j-1} = a_{j-2} = \dots = a_0$, where $b_1 = a_{j-1}$, $b_0 = a_j$, where $(b_0)^j$ denotes j consecutive b_0 s.

For convenience, $a_{t-1}a_{t-2}\dots a_1a_0$ is called the labelling of a node in $V(WK_{(d,t)})$. Fig. 1 is an illustration of $WK_{(3,3)}$.

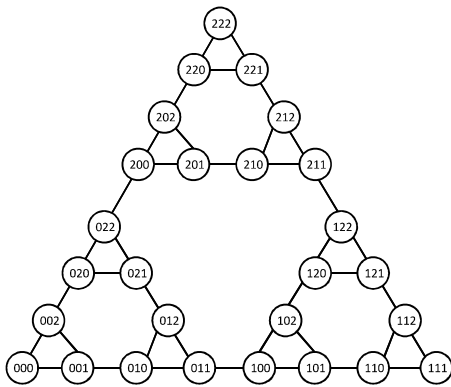


Figure 1: The structure of $WK_{(3,3)}$.

Definition 2 [7] For $d \geq 1$, $t \geq 1$ and $L \geq 0$, the radix- t WK -recursive pyramid network with amplitude d and level L , $WKP_{(d,t,L)}$, consists of a set of nodes $V(WKP_{(d,t,L)}) = \{(k, a_{tk-1}a_{tk-2}\dots a_1a_0) \mid$

$0 \leq k \leq L, 0 \leq a_i < d, 0 \leq i \leq tk-1\}$ arranged in L levels of WK -recursive network. A node is addressed as $(k, a_{tk-1}a_{tk-2}\dots a_1a_0)$ and is said to be a node at level k . The part $a_{tk-1}a_{tk-2}\dots a_1a_0$ of the address determines the address of a node within the layer k of the WK -recursive network. The nodes at level k form a network $WK_{(d,tk)}$, i.e., a node with the address $(k, a_{tk-1}a_{tk-2}\dots a_1a_0)$ placed at level k of the $WK_{(d,tk)}$ network, is connected to adjacent nodes as defined in Definition 1. This node is also connected to nodes $(k+1, a_{tk-1}a_{tk-2}\dots a_1a_0(0)^t), (k+1, a_{tk-1}a_{tk-2}\dots a_1a_0(1)^t), \dots, (k+1, a_{tk-1}a_{tk-2}\dots a_1a_0(d-1)^t)$ in level $k+1$, as childnodes, and to node $(k-1, a_{tk-1}a_{tk-2}\dots a_{t+1}a_t)$ as the parent, if $a_{t-1} = a_{t-2} = \dots = a_0$.

Fig. 2 shows $WKP_{(4,2,2)}$. In this paper, we consider the variable $t = 1$ in $WKP_{(d,t,L)}$, i.e., this is a simple version of $WKP_{(d,t,L)}$. Because the L th level of this topology is $WK_{(d,L)}$, we denote the simpler structure by $WKP_{(d,L)}$, and we give the definition as follows.

Definition 3 [10] For $d \geq 1$ and $L \geq 0$, the basic WK -recursive pyramid network, $WKP_{(d,L)}$ with amplitude d and level L , consists of a set of nodes $V(WKP_{(d,L)}) = \{(k, a_{k-1}a_{k-2}\dots a_1a_0) \mid 0 \leq k \leq L, 0 \leq a_i < d, 0 \leq i \leq k-1\}$ arranged in L levels of WK -recursive network. A node is addressed as $(k, a_{k-1}a_{k-2}\dots a_1a_0)$ and is said to be a node at level k . (0) is called apex at level 0. The part $a_{k-1}a_{k-2}\dots a_1a_0$ of the address determines the address of a node within the layer k of the WK -recursive network. The nodes at level k form a network of $WK_{(d,k)}$, i.e., a node v with the address $(k, a_{k-1}a_{k-2}\dots a_1a_0)$ placed at level k of the $WK_{(d,k)}$ network, is connected to adjacent nodes as defined in Definition 1. This node is also connected to nodes $(k+1, a_{k-1}a_{k-2}\dots a_1a_0(0)), (k+1, a_{k-1}a_{k-2}\dots a_1a_0(1)), \dots, (k+1, a_{k-1}a_{k-2}\dots a_1a_0(d-1))$ in level $k+1$, as childnodes, and to node $(k-1, a_{k-1}a_{k-2}, \dots, a_1)$ as the parent, $p(v)$. Conversely, v is called a child of $p(v)$. Note that, the node $p(v)$ has d children, and v is one of them.

Fig. 3 shows $WKP_{(3,2)}$. For convenient, a level k of $WKP_{(d,L)}$ is denoted by $WK_{(d,k)}^*$ in this paper. That is, for $1 \leq k \leq L$, $V_k = \{(k, x_{k-1}x_{k-2}\dots x_1x_0) \mid 0 \leq x_i < d \text{ and } 0 \leq i \leq k-1\}$, and $WK_{(d,k)}^*$ is the subgraph of $WKP_{(d,L)}$ induced by V_k .

Lemma 1 [4] [12] $WK_{(d,t)}$ is Hamiltonian, for $t \geq 1$ and $d \geq 3$.

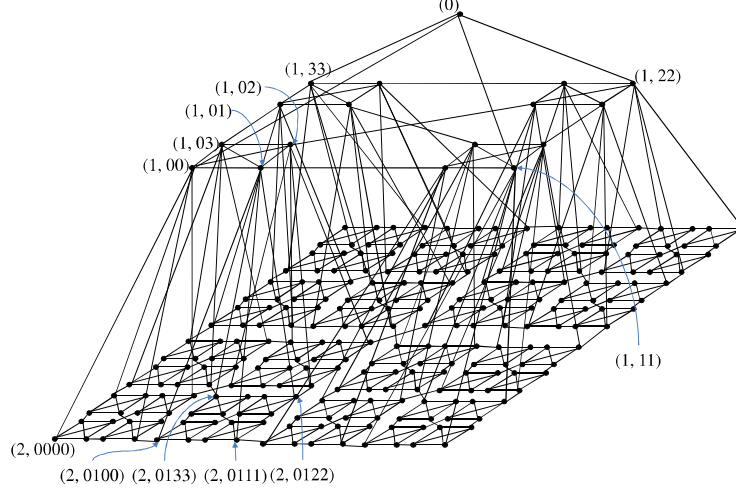


Figure 2: The structure of $WKP_{(4,2,2)}$.

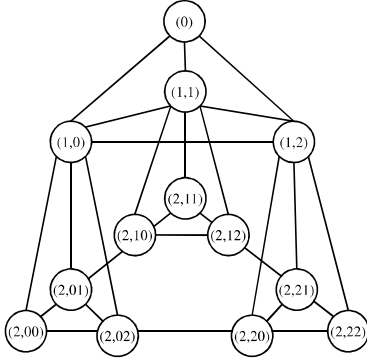


Figure 3: The structure of $WKP_{(3,2)}$.

Lemma 2 [4] For $d \geq 4$, $WK_{(d,t)}$ is Hamiltonian connected.

Theorem 1 [10] For $d \geq 3$ and $L \geq 1$, $WKP_{(d,L)}$ is Hamiltonian.

Property 1 The order $N_{(d,L)}$ of $WKP_{(d,L)}$ is $(d^{L+1} - 1)/(d - 1)$.

In this paper, we define the *outline graph* similarly to [5]. The outline graph of $WKP_{(d,L)}$, denoted by $OG(WKP_{(d,L)})$, is to take the subgraph a) the apex and all nodes in level 1, and b) each $WK_{(d,1)}$ subnetwork in level 2 to d as a supernode. Any two supernodes are adjacent if there is an edge between the original two subgraphs of

these two supernodes. Because the definition of $WKP_{(d,L)}$, we have the following property.

Property 2 $OG(WKP_{(d,L)})$ are isomorphic to $WKP_{(d,L-1)}$ for $d \geq 2$ and $L \geq 1$.

As illustrated in Fig. 4, $OG(WKP_{(3,2)})$ is $WKP_{(3,1)}$.

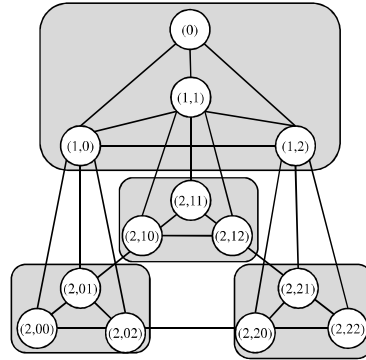


Figure 4: The outline graph of $WKP_{(3,2)}$.

Property 3 [10] The diameter of a $WKP_{(d,L)}$ is $2^L - 1$.

For other familiar definitions and notations, please refer to [11].

3 Pancyclic and m -pancycle-connectivity

In this section, we will discuss the main result of this paper. First, we need to show the following lemma.

Lemma 3 *If there exists a cycle of length q in $OG(WKP_{(d,L)})$, $WKP_{(d,L)}$ embeds cycles of all lengths ranging from $2q$ to dq .*

Proof. Because all supernodes are either K_d or K_{d+1} , there is a path P with length $|P|$ between any two nodes in any supernode, for any $1 \leq |P| \leq d$.

Suppose that there is a cycle of length q , denoted by $\langle V_0^*, V_1^*, V_2^*, \dots, V_{q-1}^*, V_0^* \rangle$, in $OG(WKP_{(d,L)})$ as illustrated in Fig. 5.

We can easily find two nodes s_i and t_i in each supernode V_i^* such that $(t_i, s_{(i+1) \bmod q}) \in E(WKP_{(d,L)})$. Obviously, $\langle s_0, t_0, s_1, t_1, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$ forms a cycle of length $2q$. When we change the different sizes of paths between s_i and t_i for any $0 \leq i \leq q-1$, we construct a cycle with lengths ranging from $2q$ to dq in $WKP_{(d,L)}$. $\square$

Then, we prove the following theorem.

Theorem 2 $WKP_{(d,L)}$ is pancyclic for $d \geq 3$.

Proof. We will prove the theorem by induction on L . For $L = 1$, $WKP_{(d,1)}$ is a K_{d+1} . Clearly, it embeds cycles of all lengths ranging from 3 to $d+1$, where $d \geq 3$.

Suppose that $WKP_{(d,L)}$ is pancyclic. Consider $WKP_{(d,L+1)}$, we know that $OG(WKP_{(d,L+1)})$ is $WKP_{d,L}$ by Property 2, and $WKP_{(d,L)}$ embeds cycles of all lengths ranging from 3 to $N_{(d,L)} = (d^{k+1} - 1)/(d - 1)$ by induction hypothesis.

By Lemma 3, $WKP_{(d,L+1)}$ can embed $\{C_s | 2n \leq s \leq dn \text{ for } 3 \leq n \leq N_{(d,L)} = (d^{k+1} - 1)/(d - 1)\}$. Since $dn - 2(n + 1) = n(d - 2) - 2 \geq n - 2 \geq 1$, for $d \geq 3$ and $3 \leq n \leq N_{(d,L)} - 1$, dn is always greater than $2(n + 1)$ for $d \geq 3$ and $3 \leq n \leq N_{(d,L)} - 1$. Hence, $WKP_{(d,L+1)}$ embeds cycles of all lengths ranging from 3 to $N_{(d,L+1)} - 1$.

Beside, $d((d^{L+1} - 1)/(d - 1)) = ((d^{L+2} - 1 - (d - 1))/(d - 1)) = (d^{k+1} - 1)/(d - 1) - 1$. So, $dN_{(d,L)} = N_{(d,L+1)} - 1$. By Theorem 1, there is a Hamilton cycle in $WKP_{(d,L+1)}$. Hence, this theorem is proved. $\square$

Now, we discuss the m -pancycle-connectivity of basic WK-recursive Pyramids. To study the property of the WK-recursive Pyramid, the following lemmas are required.

Lemma 4 *If any two supernodes are in a common cycle of length q in $OG(WKP_{(d,L)})$, any two nodes of $WKP_{(d,L)}$ are in a common cycle of all lengths ranging from $2q + 2$ to dq for $d \geq 3$ and $L \geq 2$.*

Proof. Because all supernodes are either K_d or K_{d+1} , there is a path P with length $|P|$ between any two nodes in any supernode, for any $1 \leq |P| \leq d$. Let x and y be any two nodes of $WKP_{(d,L)}$. Let V_x^* and V_y^* be the supernodes that x and y are in, respectively. By the assumption, there exists a cycle, denoted by $\langle V_0^*, V_1^*, V_2^*, \dots, V_{q-1}^*, V_0^* \rangle$, of length q in $OG(WKP_{(d,L)})$, where $\{V_x^*, V_y^*\} \subseteq \{V_0^*, V_1^*, V_2^*, \dots, V_{q-1}^*\}$. Clearly, there exists one edge connecting each of two consecutive supernodes in the cycle. Hence, we can find two nodes s_i and t_i in each supernode V_i^* , where $0 \leq i \leq q-1$, such that they are incident to the edges connecting to $V_{i-1 \bmod q}^*$ and $V_{i+1 \bmod q}^*$, respectively. Hence, $\langle s_0, t_0, s_1, t_1, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$ forms a cycle of length $2q$.

Case 1. x and y are in the same supernode V_i^* .

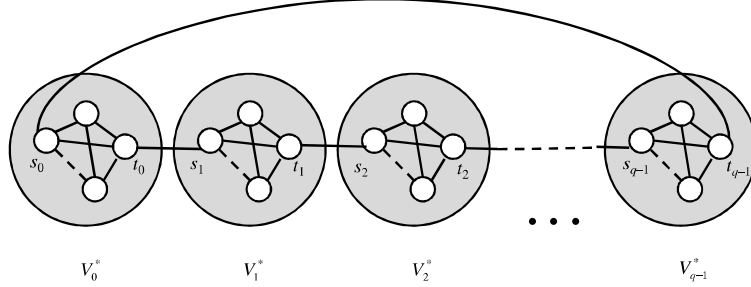
Subcase 1.1 $x, y \in \{s_i, t_i\}$. Without loss of generality, let $x = s_i$ and $y = t_i$. We can find a cycle $\langle s_0, t_0, s_1, t_1, \dots, s_i = x, t_i = y, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$. The length of this cycle is $2q$. When we change the different sizes of paths between s_i and t_i for $0 \leq i \leq q-1$, we construct cycles of all lengths ranging from $2q$ to dq .

Subcase 1.2 One of x and y is in $\{s_i, t_i\}$, and the other is not. Without loss of generality, let $x = s_i$. We can find a cycle $\langle s_0, t_0, s_1, t_1, \dots, s_i = x, y, t_i, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$. The length of this cycle is $2q + 1$. When we change the different sizes of paths between s_i and t_i for $0 \leq i \leq q-1$, we construct cycles of all lengths ranging from $2q + 1$ to dq .

Subcase 1.3 $x, y \notin \{s_i, t_i\}$. We can find a cycle $\langle s_0, t_0, s_1, t_1, \dots, s_i, x, y, t_i, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$. The length of this cycle is $2q + 2$. When we change the different sizes of paths between s_i and t_i for $0 \leq i \leq q-1$, we construct cycles of all lengths ranging from $2q + 2$ to dq .

Case 2. x and y are in supernodes V_i^* and V_j^* , respectively. Without loss of generality, we assume that $i < j$.

Subcase 2.1 $x \in \{s_i, t_i\}$ and $y \in \{s_j, t_j\}$. Without loss of generality, let $x = s_i$ and $y = s_j$. We can find a cycle $\langle s_0, t_0, s_1, t_1, \dots, s_i = x, t_i, \dots, s_j = y, t_j, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$. The length of this cycle is $2q$. When we change the different sizes of paths between s_i and t_i for $0 \leq i \leq q-1$, we construct cycles of all lengths ranging from $2q$ to dq .


 Figure 5: A cycle of length q in $\text{WKP}_{(d,L)}$.

Subcase 2.2 One of x and y is in $\{s_i, t_i, s_j, t_j\}$, and the other is not. Without loss of generality, let $x = s_i$. We can find a cycle $\langle s_0, t_0, s_1, t_1, \dots, s_i = x, t_i, \dots, s_j, y, t_j, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$. The length of this cycle is $2q + 1$. When we change the different sizes of paths between s_i and t_i for $0 \leq i \leq q - 1$, we construct cycles of all lengths ranging from $2q + 1$ to dq .

Subcase 2.3 $x \notin \{s_i, t_i\}$ and $y \notin \{s_j, t_j\}$. We can find a cycle $\langle s_0, t_0, s_1, t_1, \dots, s_i, x, t_i, \dots, s_j, y, t_j, \dots, s_{q-1}, t_{q-1}, s_0 \rangle$. The length of this cycle is $2q + 2$. When we change the different sizes of paths between s_i and t_i for $0 \leq i \leq q - 1$, we construct cycles of all lengths ranging from $2q + 2$ to dq .

Thus, x and y are in a cycle of all lengths ranging from $2q + 2$ to dq . $\square$

For convenient, we assume that S is the node set of all nodes in lever 0 and 1, the induce subgraph of S is isomorphic to the complete graph K_{d+1} . We also find paths of all lengths ranging from 1 to d between any two nodes in S for $d \geq 3$. For any two nodes a and b in S , let $P^l(a, b)$ be one (a, b) -path with length l in S .

Lemma 5 $\text{WKP}_{(d,2)}$ is 6-pancycle-connected for $d \geq 3$.

Proof. By Lemma 2, we know $\text{OG}(\text{WKP}_{(d,2)}) = \text{WKP}_{(d,1)} = K_{d+1}$. Hence, $\text{OG}(\text{WKP}_{(d,2)})$ is 3-pancycle-connected. By Lemma 4, $\text{WKP}_{(d,L+1)}$ can embed $\{C_s | 2n+2 \leq s \leq dn \text{ for } 3 \leq n \leq d+1\}$. Since $(dn+1) - (2(n+1)+2) = n(d-2) - 3 \geq n-3 \geq 0$, for $d \geq 3$ and $3 \leq n \leq d+1$, $dn+1$ is always greater than or equal to $2(n+1)+2$ for $d \geq 3$ and $3 \leq n \leq d+1$. So, we can construct cycles of all lengths ranging from 8 to $d(d+1)$.

Because $|V(\text{WKP}_{(d,2)})| = d(d+1) + 1$, we only need to prove that any two nodes of $\text{WKP}_{(d,2)}$ are in common cycles of lengths 6, 7 and $d(d+1) + 1$. By Lemma 1, we know that $V(\text{WKP}_{(d,2)})$ is a Hamilton. So, any two nodes of $\text{WKP}_{(d,2)}$ are in common cycles of length $d(d+1) + 1$. Now, we prove that any two nodes of $\text{WKP}_{(d,2)}$ are in common cycles of lengths 6 and 7. Let u and v are any two nodes of $\text{WKP}_{(d,2)}$. According to the location of the nodes, we have two cases as follows. In the following cases, we will construct two cycles C_1 and C_2 connecting u and v , such that the length of C_1 and C_2 are 6 and 7, respectively.

Case 1 One of u and v is in level 0. Without loss of generality, let $v = (0)$.

Subcase 1.1 u is in level 1. Let $u = (1, u_0)$ and $c = (u_0 + 1) \bmod d$. We can construct cycles $C_1 = \langle (0), (1, u_0), (2, u_0 u_0), (2, u_0 c), (2, c u_0), (1, c), (0) \rangle$ and $C_2 = \langle (0), (1, u_0), (2, u_0 u_0), (2, u_0 c), (2, c u_0), (2, cc), (1, c), (0) \rangle$.

Subcase 1.2 u is in level 2. Without loss of generality, let $u = (2, u_1 u_0)$ and $c = (u_0 + 1) \bmod d$. We can construct cycles $C_1 = \langle (0), (1, u_1), (2, u_1 u_0), (2, u_1 c), (2, c u_1), (1, c), (0) \rangle$ and $C_2 = \langle (0), (1, u_1), (2, u_1 u_0), (2, u_1 c), (2, c u_1), (2, cc), (1, c), (0) \rangle$.

Case 2 u and v are not in level 0.

Subcase 2.1 u and v are in level 1. Let $u = (1, u_0)$ and $v = (1, v_0)$. We can construct cycles $C_1 = \langle (0), (1, u_0), (2, u_0 u_0), (2, u_0 v_0), (2, v_0 u_0), (1, v_0), (0) \rangle$ and $C_2 = \langle (0), (1, u_0), (2, u_0 u_0), (2, u_0 v_0), (2, v_0 u_0), (2, v_0 v_0), (1, v_0), (0) \rangle$.

Subcase 2.2 u and v are in level 2. By Property 3, $d(u, v)$ is at most 3. Hence, $d(u, v)$ is possibly equal to 1, 2 or 3. For any two nodes u and v in $\text{WKP}_{(d,2)}$, we denote the shortest path between u and v by $P_{\text{WKP}_{(d,2)}}^*(u, v)$. If $p(u) \neq p(v)$, we construct a cy-

cle $C_1 = \langle u, P_{WK_{(d,2)}^*}(u, v), v, p(v), P^{4-d(u,v)}(p(v), p(u)), p(v), u \rangle$ and $C_2 = \langle u, P_{WK_{(d,2)}^*}(u, v), v, p(v), P^{5-d(u,v)}(p(v), p(u)), p(u), u \rangle$. If $p(u) = p(v)$, we assume that $u = (2, u_1 u_0)$, $v = (2, u_1 v_0)$ and $p(u) = p(v) = (1, u_1)$. Because u and v are distinct nodes, at least one is not $(2, u_1 u_1)$. Without loss of generality, we assume $u_1 \neq u_0$. We construct a cycle $C_1 = \langle v, u, (2, u_1 u_0), (1, u_1), P^2((1, u_1), p(v)), p(v), v \rangle$ and $C_2 = \langle v, u, (2, u_1 u_0), (1, u_1), P^3((1, u_1), p(v)), p(v), v \rangle$.

Subcase 2.3 One of u and v is in level 1, and the other is in level 2. Without loss of generality, say u is in level 1. Consider a child node of u which is not equal to v , denoted by u' . We construct C_1 and C_2 similar to Subcase 2.2 for u' and v . Note that C_1 and C_2 will contain u and v as our desire.

Thus, we prove $WKP_{(d,2)}$ is 6-pancycle-connected for $d \geq 3$. $\square$

Theorem 3 $WKP_{(d,L)}$ is $(2^{L+1} - 2)$ -pancycle-connected for $d \geq 3$ and $L \geq 2$.

Proof. We will prove the theorem by induction on L . For $L = 2$, the theorem is true by Lemma 5. Suppose that any two nodes of $WKP_{(d,L)}$ are in a common cycle of all lengths ranging from $2^{(L+1)} - 2$ to $N_{(d,L)}$, where $d \geq 3$. Consider $WKP_{(d,L+1)}$, we know that $OG(WKP_{(d,L+1)})$ is isomorphic to $WKP_{(d,L)}$ by Property 2. By induction hypothesis and Lemma 4, we know that any two nodes of $WKP_{(d,L+1)}$ are in $\{C_s | 2n + 2 \leq s \leq dn \text{ for } 2^{(L+1)} - 2 \leq n \leq N_{(d,L)} = (d^{L+1} - 1)/(d - 1)\}$. Since $dn - (2(n+1) + 2) = n(d-2) - 4 \geq n - 4 \geq 0$ for $d \geq 3$ and $n \geq 2^{L+1} - 1 \geq 6$, nd is always greater than or equal to $2(n+1) - 2$ for $d \geq 3$ and $2^{(L+1)} - 2 \leq n \leq N_{(d,L)}$. That is, any two nodes of $WKP_{(d,L+1)}$ are in $\{C_s | 2^{(L+2)} - 2 \leq s \leq dN_{(d,L)}\}$ for $d \geq 3$. Beside, $d((d^{L+1} - 1)/(d - 1)) = (d^{L+2} - 1 - (d - 1))/(d - 1) = (d^{L+2} - 1)/(d - 1) - 1 = N_{(d,L+1)} - 1$. So, $dN_{(d,L)} = N_{(d,L+1)} - 1$. By Therome 1, there is a Hamilton cycle in $WKP_{(d,L+1)}$. Hence, this therom is proved. $\square$

Lemma 6 $WKP_{(d,L)}$ is not panconnected for $d \geq 2$ and $L \geq 2$.

Proof. Consider the pair of nodes $(2,01)$ and $(2,10)$ in $WKP_{(d,L)}$ for $d \geq 2$ and $L \geq 2$. By the definition, the distance between $(2,01)$ and $(2,10)$ is 1, and there is no common neighbor in $WKP_{(d,L)}$. Obviously, there exists no path of length 2 between $(2,01)$ and $(2,10)$. $\square$

4 Conclusion

In this paper, we show that the basic WK-Recursive Pyramid with amplitude d and level L is pancyclic and $(2^{L+1} - 2)$ -pancycle-connected for $d \geq 3$. That is, there exist cycles of all lengths ranging from 3 to $|V(WKP_{d,L})|$ and for any two nodes are in a common cycle of all lengths ranging from $(2^{L+1} - 2)$ to $|V(WKP_{d,L})|$ in WK-Recursive Pyramid with amplitude greater than or equal to 3. We also show that the basic WK-Recursive Pyramid with $L \geq 2$ is not panconnected. In the future, we will try to find an interger x , such that the basic WK-Recursive Pyramid $WK_{(d,L)}$ is x -panconnected.

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References

- [1] T. Araki and Y. Shibata. Pancyclicity of recursive circulant graphs. *Information Processing Letters*, 81(4):187–190, 2002.
- [2] F. Buckley and F. Harary. *Distance in Graphs*. Reading, Mass.: Addison-Wesley, 1990.
- [3] J.-M. Chang, J.-S. Yang, J.-S. Yang, Y.-L. Wang, and Y. Cheng. Pan-connectivity, fault-tolerant hamiltonicity and hamiltonianconnectivity in alternating group graphs in alternating group graphs. *Networks*, 44(4):302–310, 2004.
- [4] G.H. Chen and D.R. Duh. Topological properties, communication, and computation on wk-recursive network. *Networks*, 24(6):303–317, 1994.
- [5] J.-F. Fang, Y.-R. Wang, and H.-L. Huang. The m -pancycle-connectivity of a wk-recursive network. *Information Sciences*, 177:5611–5619, 2007.
- [6] R. Fernandes. Recursive interconnection networks for multicomputer networks. *Proceedings of the 1992 International Conference on Parallel Processing*, I:76–79, 1992.

- [7] R. Fernandes and A. Kanevsky. Hierarchical wk-recursive topologies for multicomputer systems. *Proceedings of the 1993 International Conference on Parallel Processing*, I:315–318, 1993.
- [8] J.-S. Fu. Hamiltonicity of the wk-recursive network with and without faulty nodes. *IEEE Transactions on Parallel and Distributed Systems*, 16(9):853–865, Sep 2005.
- [9] S. Razavi and H. Sarbazi-Azad. The triangular pyramid: Routing and topological properties. *Information Sciences*, 180:2328–2339, 2010.
- [10] Yi-Chun Wang and Justie Su-Tzu Juan. The hamiltonicity of wk-recursive pyramid. In *Algorithms and Architectures for Parallel Processing*, volume 7440 of *Lecture Notes in Computer Science*, pages 90–99. Springer, 2012.
- [11] J. Xu. *Theory and Application of Graphs*. Kluwer Academic Publishers, 2003.
- [12] Dejan Zivkovic. Hamiltonianicity of the tower of hanoi problem. *Univ. Beograd. Publ. Elektrotehn. Fak. Ser. Mat.*, 17:31–37, 2006.